

Homework 1

Due: August 30, 2026

The goal of this homework is to get familiar with ITCH and EBS high-frequency order and transaction data. Choose 2 stocks that differ in their size (market capitalization). Use NASDAQ tickers. Use the currency pairs: EUR/JPY, EUR/USD, and USD/JPY.

For one trading day (9:30 to 4) calculate descriptive summary statistics for the following variables:

- a. dollar trading volume per minute
- b. number of trades and number of orders (orders only for NASDAQ ITCH) per minute
- c. open, close, high, and low prices
- d. VWAP per minute
- e. BBO spread and depth per minute
- f. depth at twice that day's average spread (it will be zero when the spread is greater than twice its average) per minute
- g. 5-second price impact (regress 5-second midpoint quote return on current trade sign, -1 or +1) per minute
- h. Calculate one second and one-minute midquote and transaction price series (take the last quote price and transaction price in each interval).
- i. Compute the difference (log-returns), $\log(p_t) - \log(p_{t-1})$, for both midquote prices and transaction prices. These are the midquote return and transaction return series.
- j. Compute the realized variance for both midquote returns and transaction returns.
- k. Compute the autocorrelation for both midquote returns and transaction returns using either the Box-Pierce test (function `Box.test` from package `stats`) or ACF (function `pacf`).
- l. For the currency pairs for the one-second series, examine the occurrence of triangular arbitrage opportunities in terms of their frequency, duration, and quantity that can be traded. The other variables should be calculated for both stocks and currencies.

Provide a table of the summary statistics.

Construct daily plots for the liquidity measures (spread, depth, and price impact) that change throughout the day for each 5min interval for each day, i.e., a plot for each variable each day where the x-axis is 9:30-9:35, 9:35-9:40, ... , 3:55-4:00. For the 1-minute transaction-based log-return series, plot variance and the first autocorrelation (one lag) for each 30min interval, i.e., 9:30-10:00, 10:00-10:30, ... , 3:30-4:00.

For the auto-correlations, make a plot with the default number of lags or choose a lag length that shows the autocorrelation becoming and staying zero, e.g., 10 lags. Choose up to 4 of the more interesting graphs. Document interesting observations that you find and why these may exist. Code need to be shared in Datalore, and a report should be submitted via bCourses.

Document interesting observations that you find and relate these to your findings in Homework 1. The discussion based on the economics from class is an important aspect of this.